Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

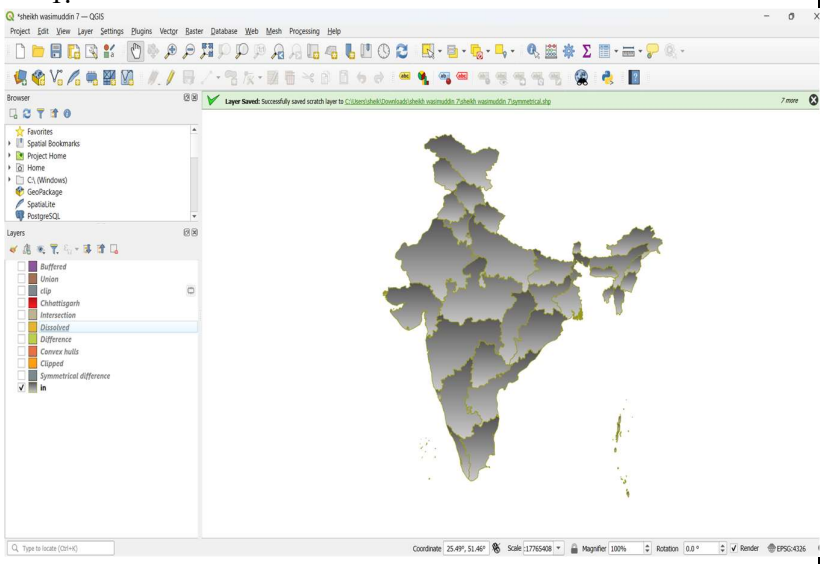
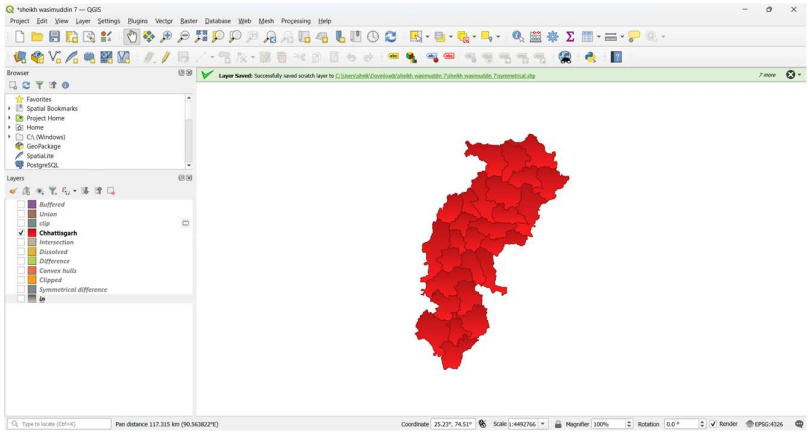
"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**08**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Implement Geo-processing Tools(Clip, Buffer, Difference, Union, Intersection and Dissolve) on various vector data by using model building in QGIS.
Problem Definition	To apply various geo-processing tools using model building in QGIS on the Chhattisgarh shapefile for spatial analysis and data visualization.
Theory (100 words)	Geo-processing tools in QGIS are essential spatial analysis operations used to manipulate and analyze vector data layers such as points, lines, and polygons. These tools help derive meaningful information by combining, comparing, or modifying spatial features. Clip extracts features from one layer based on the boundary of another layer, keeping only the portions that fall within the clipping layer. Buffer creates a zone of a specified distance around spatial features, useful for proximity analysis such as identifying areas within a certain distance from roads or rivers. Difference removes the overlapping portion of one layer from another, producing the parts that do not intersect. Union combines two or more layers into one, preserving all features and their attributes from both layers. Intersection extracts the overlapping area between two layers, keeping only the common regions and merging their attribute information. Finally, Dissolve merges adjacent polygons that share the same attribute value, simplifying the dataset and reducing redundancy. Using Model Builder in QGIS, these operations can be automated into a single workflow, allowing users to link multiple geoprocessing tools in sequence. This helps in efficiently executing complex spatial analyses without repeating manual steps, making the process faster, reproducible, and user-friendly.

Procedure and Execution (100 Words)	Implementation Steps: 1. Download India and Chhattisgarh Shapefile 2. Import it as vector layers 3. Go to Processing → Model Designer 4. Select inputs as vector and algorithm as clipped, Buffer Difference, Dissolve, Intersection and Union 5. Select Input .shp 6. Run the Algorithms And Your output Shape file will be visible on Qgis. 1.  2. 

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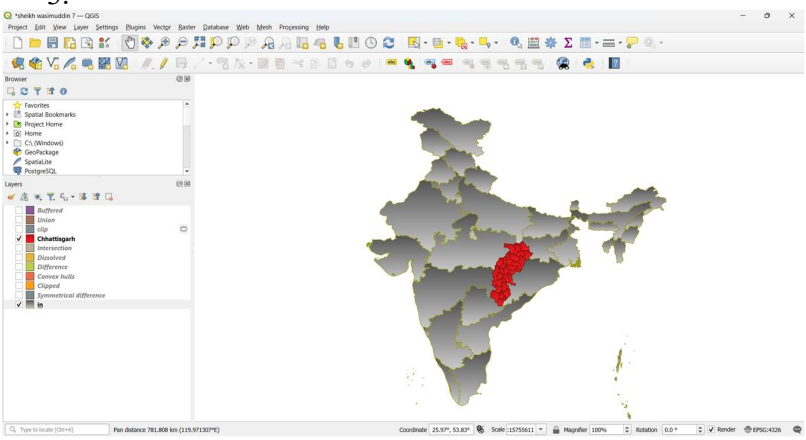

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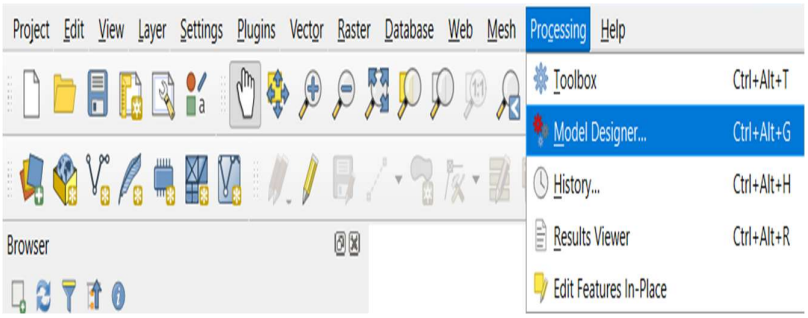
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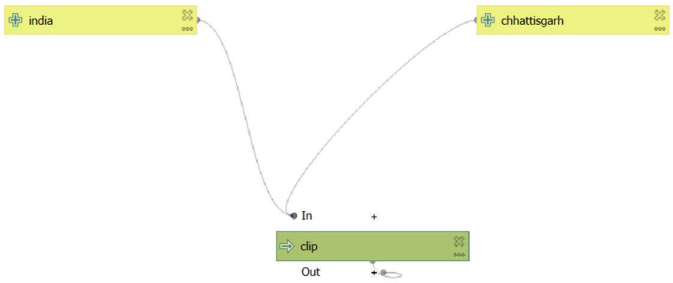
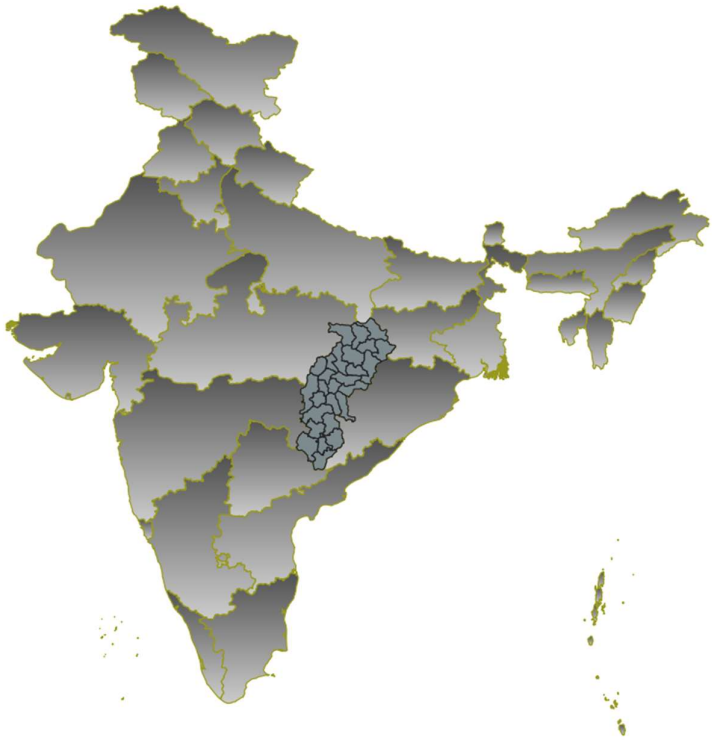
History... Ctrl+Alt+H
Results Viewer Ctrl+Alt+R
Edit Features In-Place
Browser

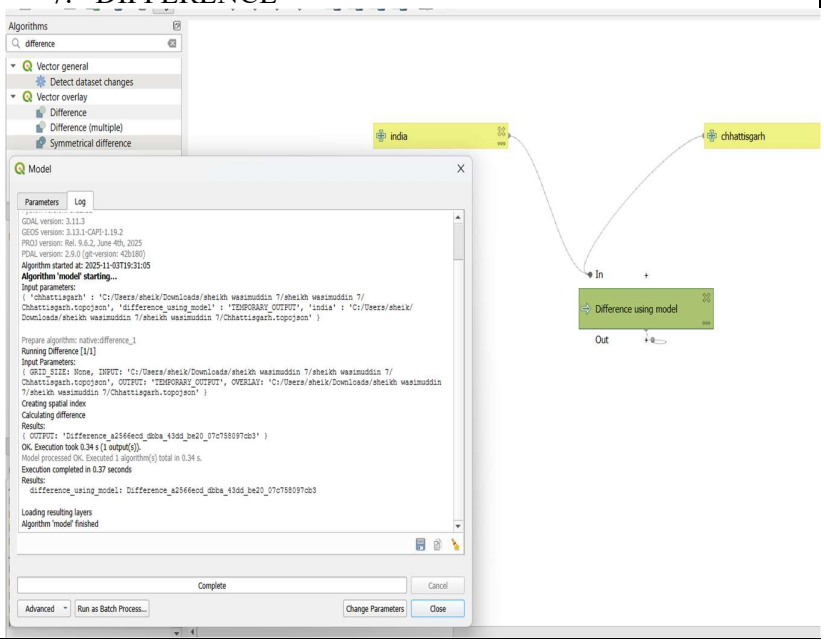
5. CLIP

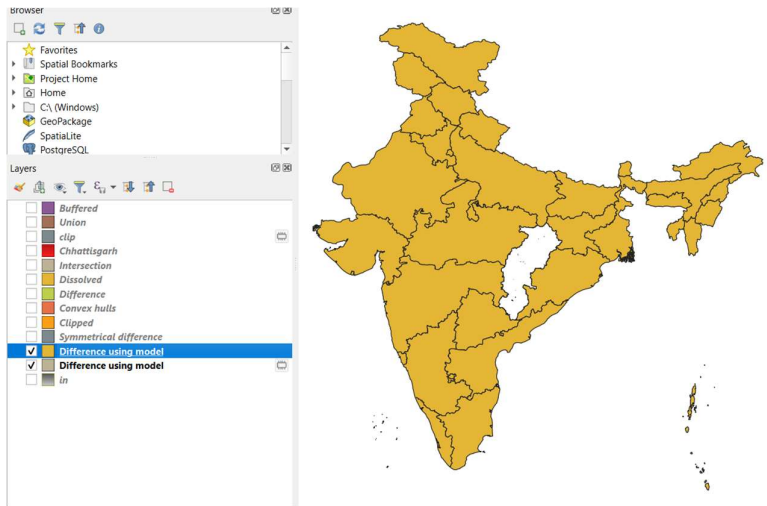

india
chhattisgarh
In
clip
Out

6.



7. DIFFERENCE

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SpatiaLite
PostgreSQL
Layers
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Chhattisgarh
Intersection
Dissolved
Difference
Convex hulls
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Symmetrical difference
Difference using model
Difference using model
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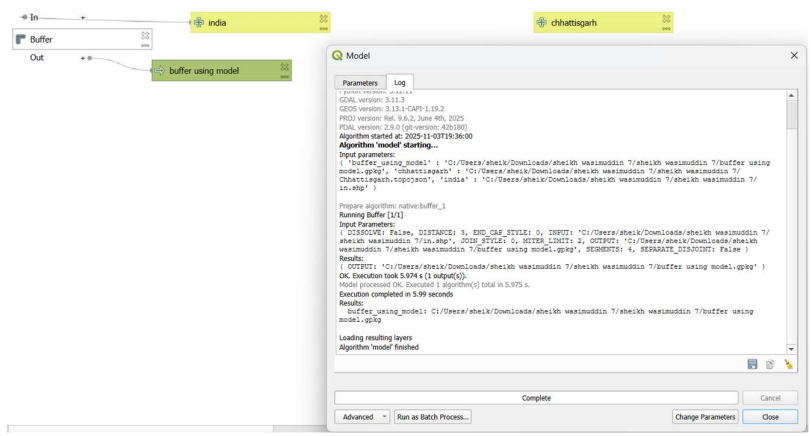
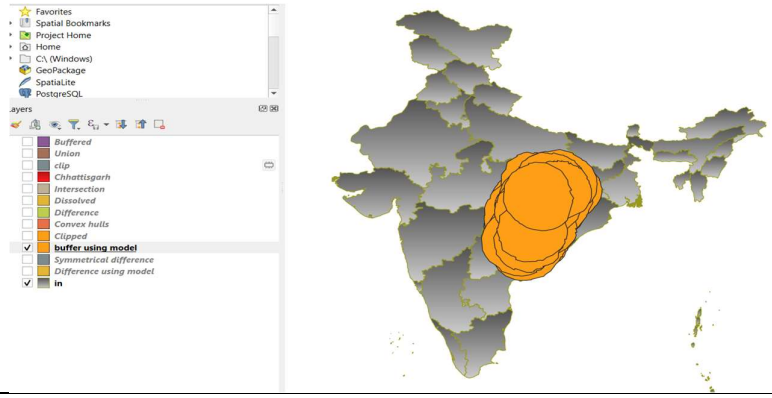
8. BUFFER


In
Buffer
Out
india
chhattisgarh
buffer using model
Model
Parameters
Log
GDAL version: 3.11.3
GEOS version: 3.13.1-CAPI-1.19.2
PROJ version: Rel. 9.6.2, June 4th, 2025
PDAL version: 2.9.0 (git-version: 42b180)
Algorithm started at: 2025-11-03T19:36:00
Algorithm 'model' starting...
Input parameters:
{ 'buffer_using_model' : 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/buffer using model.gpkg', 'chhattisgarh' : 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/Chhattisgarh.topojson', 'india' : 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/in.shp' }
Prepare algorithm: native:buffer_1
Running Buffer [1/1]
Input Parameters:
{ DISSOLVE: False, DISTANCE: 3, END_CAP_STYLE: 0, INPUT: 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/in.shp', JOIN_STYLE: 0, MITER_LIMIT: 2, OUTPUT: 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/buffer using model.gpkg', SEGMENTS: 4, SEPARATE_DISJOINT: False }
Results:
{ OUTPUT: 'C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/buffer using model.gpkg' }
OK. Execution took 5.974 s (1 output(s)).
Model processed OK. Executed 1 algorithm(s) total in 5.975 s.
Execution completed in 5.99 seconds
Results:
buffer_using_model: C:/Users/sheik/Downloads/sheikh wasimuddin 7/sheikh wasimuddin 7/buffer using model.gpkg
Loading resulting layers
Algorithm 'model' finished
Complete
Cancel
Advanced
Run as Batch Process...
Change Parameters
Close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Layers
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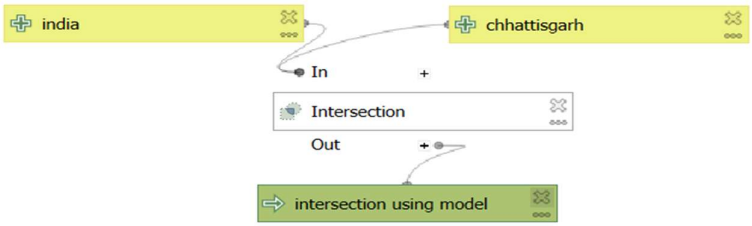
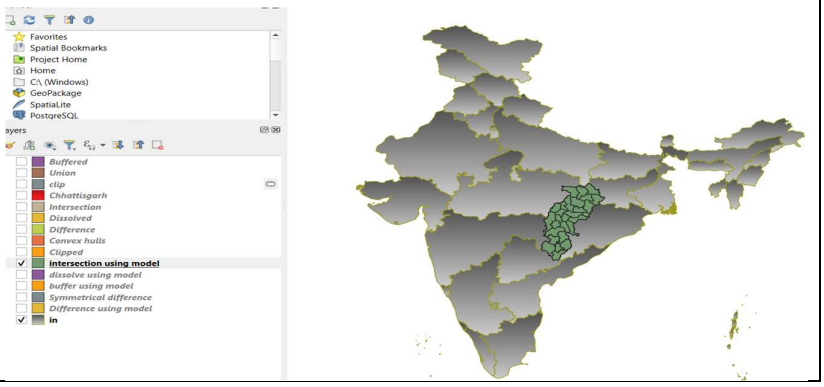
Union
clip
Chhattisgarh
Intersection
Dissolved
Difference
Convex hulls
Clipped
buffer using model
Symmetrical difference
Difference using model
in

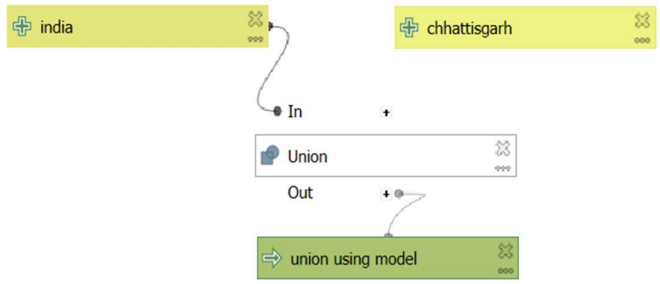
9. DISSOLVE

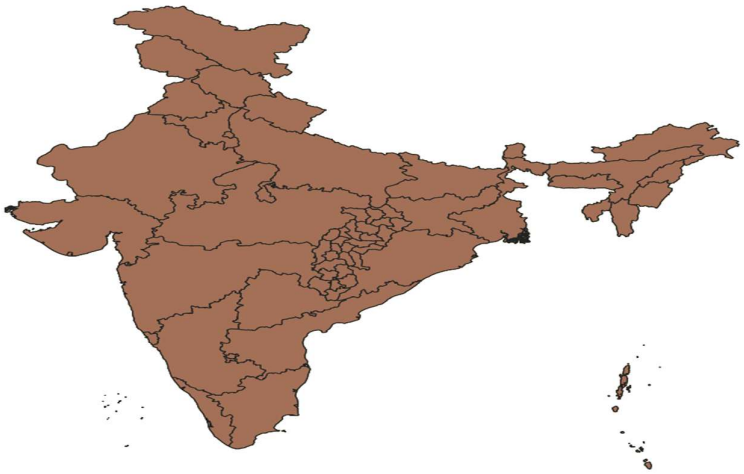
10. INTERSECTION



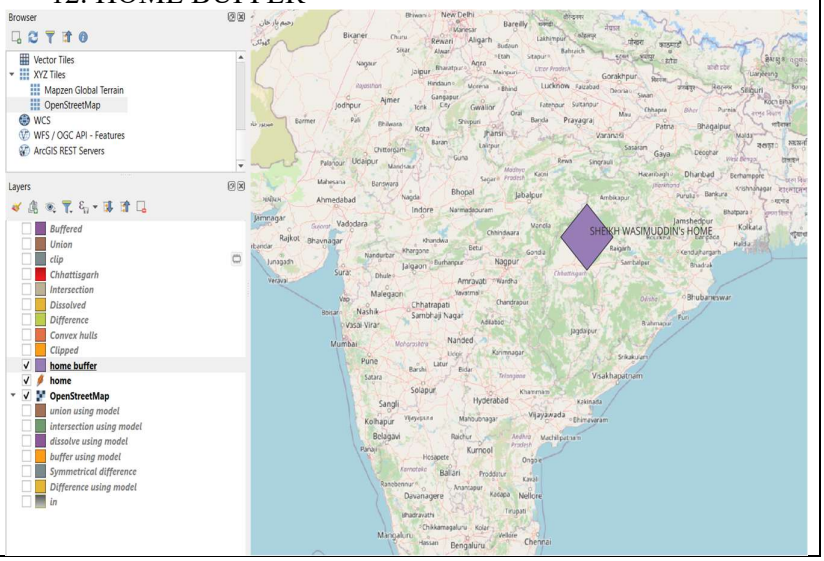


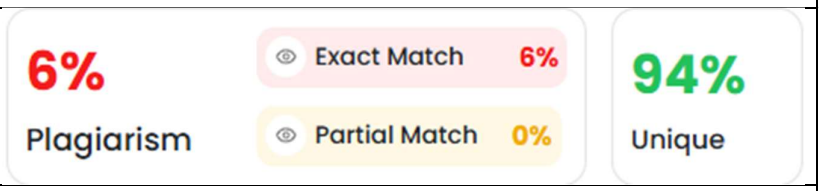
11. UNION





12. HOME BUFFER



Output Analysis	Using QGIS, a model was created to apply different geo-processing tools on the Chhattisgarh shapefile. The Clip tool was used to extract only the features within Chhattisgarh's boundary, making the map cleaner and more region-specific. The Buffer tool created zones around rivers, roads, and urban areas to analyze regions suitable for development and environmental management. The Difference tool identified areas outside these buffered zones, helping to distinguish unimpacted regions. The Union and Intersection tools were applied to combine multiple vector layers and analyze overlapping regions such as forest and mining zones. Finally, the Dissolve tool merged polygons with similar attributes, simplifying the dataset for better visualization. Overall, the model streamlined the spatial analysis process, improved efficiency, and provided clear insights into Chhattisgarh's land use and geographic patterns.
Conclusion	The implementation of various geo-processing tools such as Clip, Buffer, Difference, Union, Intersection, and Dissolve on the Chhattisgarh shapefile using model building in QGIS successfully demonstrated the power of spatial data analysis and automation. By using model building, the workflow became more efficient, repeatable, and organized. The process helped extract meaningful geographic information, identify spatial relationships, and visualize patterns such as buffer zones, overlapping regions, and merged land areas. Overall, this practical enhanced the understanding of spatial analysis techniques and provided valuable insights into Chhattisgarh's geographical and developmental features.
Plag Report (Similarity index < 12%)	
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